

COMPUTER

GRAPHICS

BCA 5TH Semester

by -

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DEPARTMENT OF COMPUTER APPLICATION

CMP DEGREE COLLEGE

UNIT - I

Introduction

Introduction of Computer graphics

- Computer graphics is an art of drawing pictures on computer screens with the help of programming.
- It involves computations, creation, and manipulation of data. In other words, we can say that computer graphics is a rendering tool for the generation and manipulation of images.
- Computer Graphics including digital images, animations, and interactive graphics used in various sectors such as entertainment, education, scientific visualization, and virtual reality. Computer Graphics can be used in UI_design, rendering, geometric objects, animation, and many more.

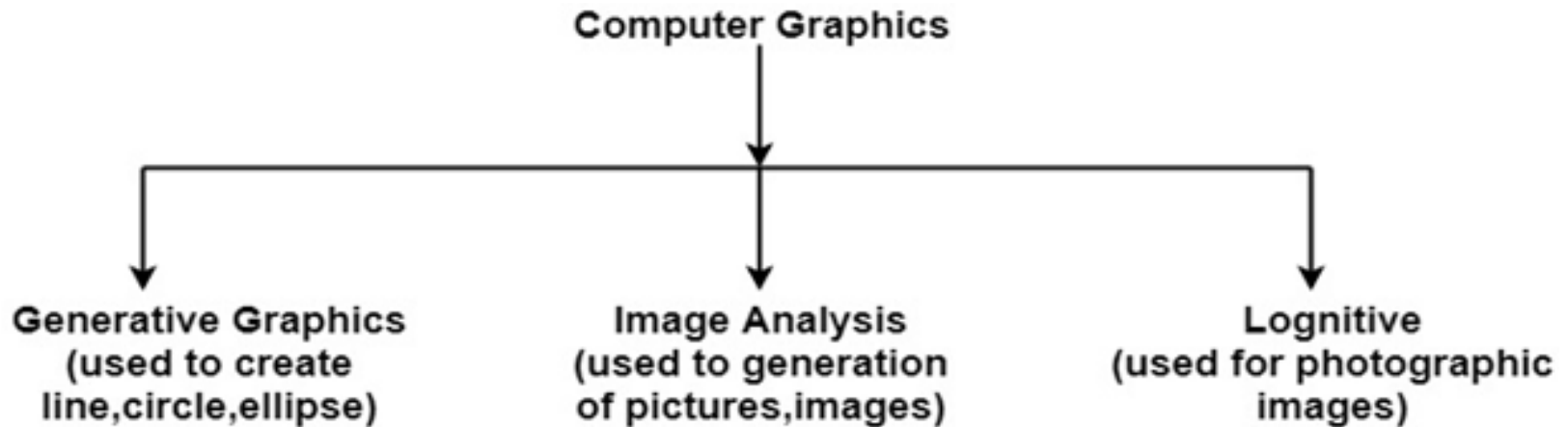
Introduction of Computer graphics

- Computer graphics are related to all activities where static and animated pictures and images are created, processed and manipulated using specialized hardware and computer software
- Data in the form of images is easy to understand and interpret. Hence the use of computers in producing static and animated images has revolutionized video games and movies production.
- Manipulating images needs powerful tools that have to process and store large amount of data. Computers offer both advantages of very high speed of operations and large capacity of data storage hence their widespread use in generating and processing graphics.

History of Computer graphics

- John Whitney was regarded as a pioneer and one of the early computer animators throughout the 1940s and 1950s.
- Early in the 1960s, William Alan Fetter, a graphic designer for Boeing America, became familiar with the phrase "computer graphics." With the introduction of graphics hardware, the phrase "computer graphics" was created.
- Early developments included simple line drawings and progressed to complex 3D models and animations.
- Computer graphics come in a variety of forms, each of which has a specific application.

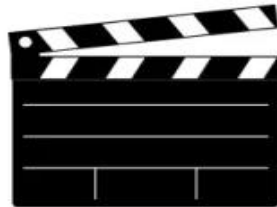
Application of Computer graphics



Applications of Computer Graphics



Education



Entertainment

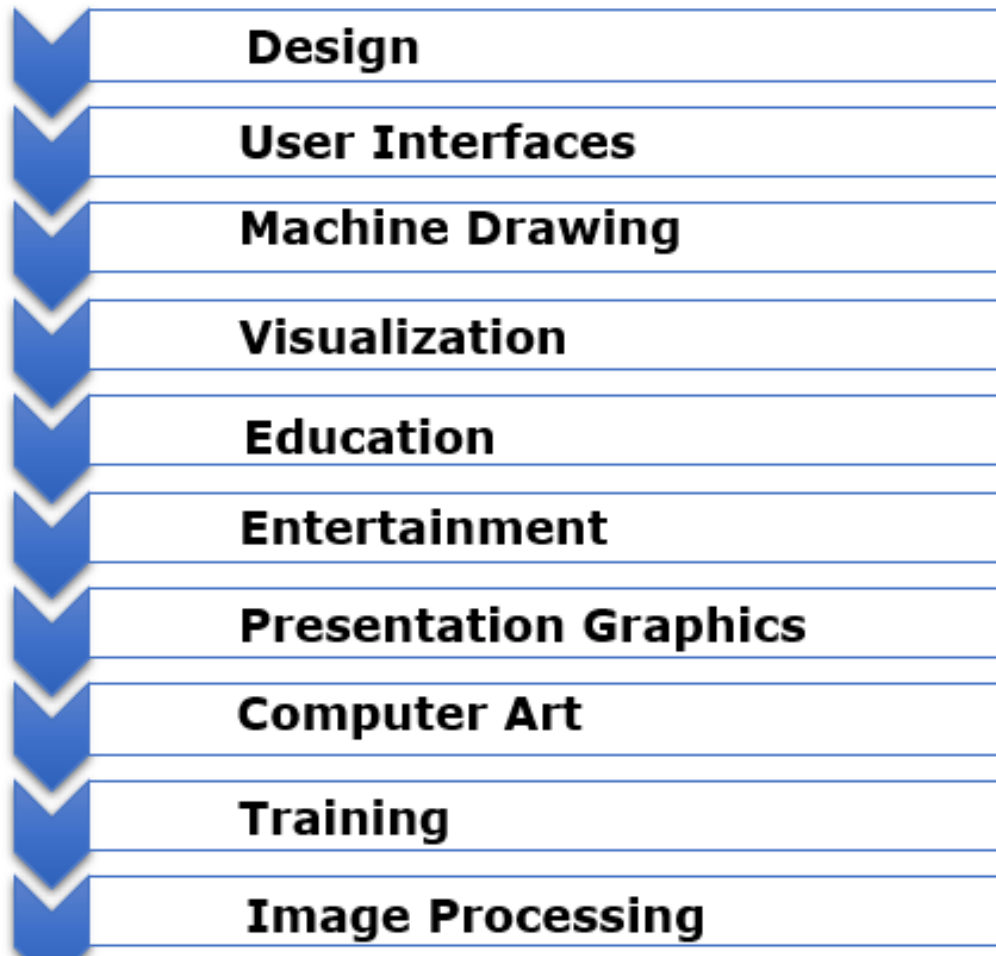


Training



Presentation

Application of Computer graphics



Application of Computer graphics

- **Film and Animation:** Creating stunning visual effects, character animation, and entire worlds.
- **Video Games:** Developing immersive gaming experiences with realistic graphics and animations.
- **Virtual Reality (VR) and Augmented Reality (AR):** Creating interactive and immersive environments.
- **Product Design:** Designing and visualizing products before physical prototypes are created.
- **Architecture and Interior Design:** Creating virtual walkthroughs and presentations of buildings and spaces.
- **Engineering and Simulation:** Visualizing complex data and simulating physical phenomena.

Application of Computer graphics

- **Medical Imaging:** Creating 3D models of human anatomy for diagnosis and treatment planning.
- **Simulation:** Simulating physical and biological systems for research and development.
- **Advertising:** Creating eye-catching visuals and animations for commercials and print media.
- **User Interface Design:** Designing intuitive and visually appealing user interfaces for software applications.
- **Data Visualization:** Creating informative and engaging visualizations of complex data sets.
- **E-learning:** Developing interactive and engaging educational materials.

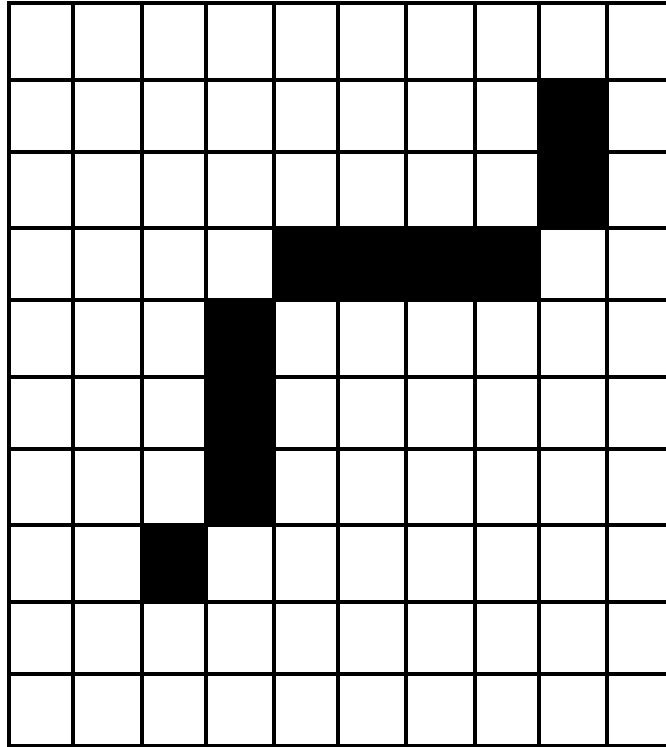
Application of Computer graphics

- **Virtual Labs:** Providing virtual access to scientific equipment and experiments.
- **Computer-Aided Manufacturing (CAM):** Generating instructions for manufacturing processes.
- **Geographic Information Systems (GIS):** Creating maps and visualizations of geographic data.
- **Image Processing:** Enhancing and manipulating digital images.
- **Computer-Aided Design (CAD):** In industries such as architecture and automotive engineering, computer-aided drawing is essential for designing buildings, vehicles, and aircraft.

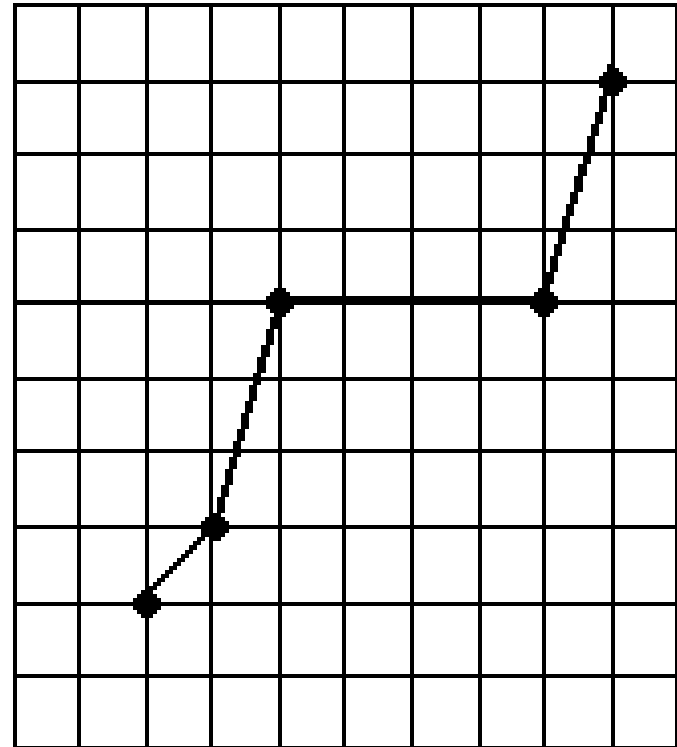
Raster & Vector Images

- A raster or bitmap image uses a grid of individual dots called pixels where each pixel is a small colored square. When an image is scanned, the image is converted to a collection of pixels called a raster image. This conversion make it easy to edit and process these images. JPEG and GIF images are examples of raster images.
- Images in vector graphics are basically points connected by lines of various shapes. Two points and a straight path between them, for example, are used to describe a segment AB but more points and more complex paths are needed for more complicated shapes.

Raster & Vector Images



Raster view



Vector view

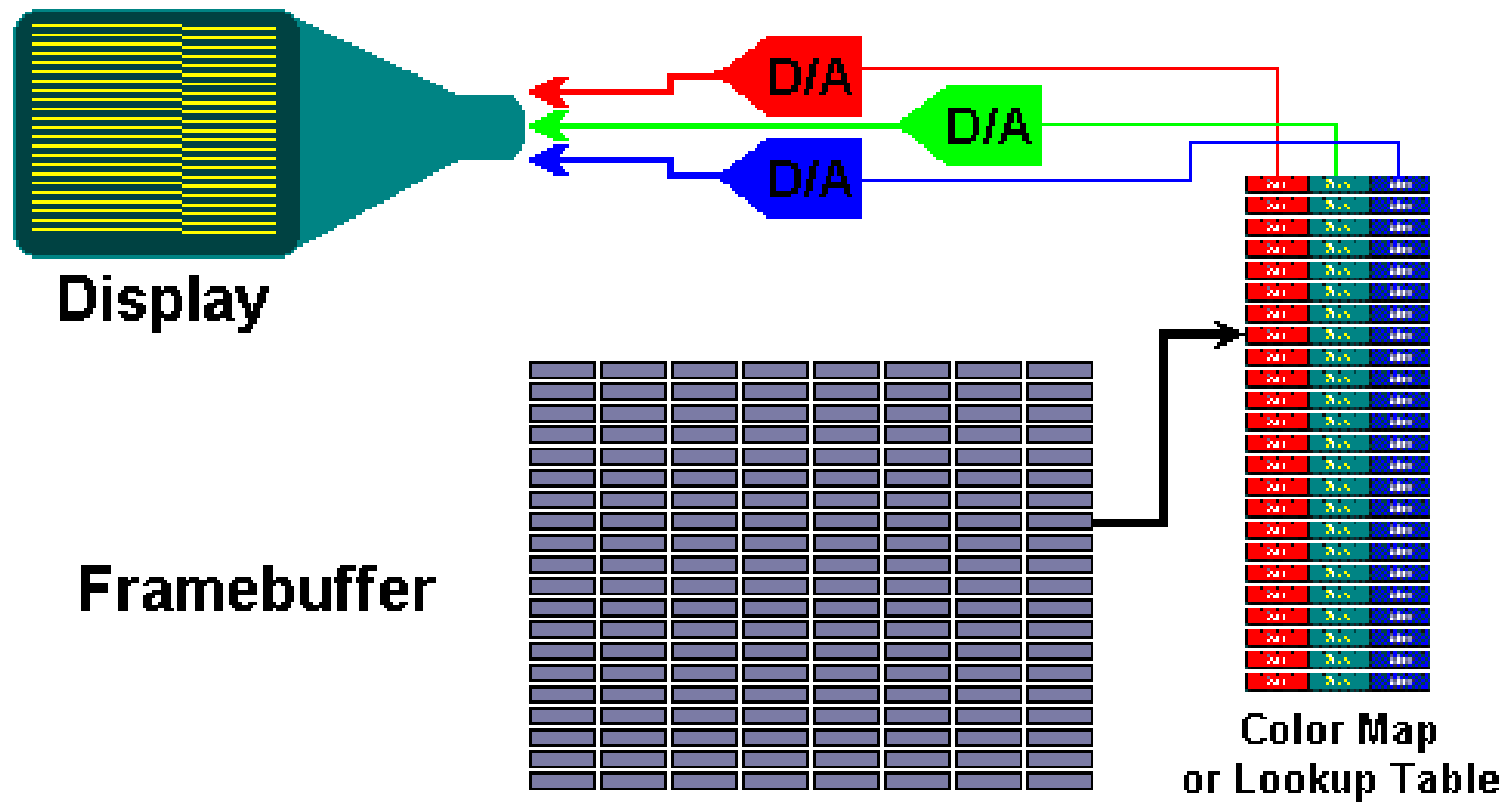
Raster & Vector Images

Raster Graphics	Vector Graphics
They are composed of pixels.	They are composed of paths.
In Raster Graphics, refresh process is independent of the complexity of the image.	Vector displays flicker when the number of primitives in the image become too large.
Graphic primitives are specified in terms of end points and must be scan converted into corresponding pixels.	Scan conversion is not required.
Raster graphics can draw mathematical curves, polygons and boundaries of curved primitives only by pixel approximation.	Vector graphics draw continuous and smooth lines.
Raster graphics cost less.	Vector graphics cost more as compared to raster graphics.
They occupy more space which depends on image quality.	They occupy less space.
File extensions: .BMP, .TIF, .GIF, .JPG	File Extensions: .SVG, .EPS, .PDF, .AI, .DXF

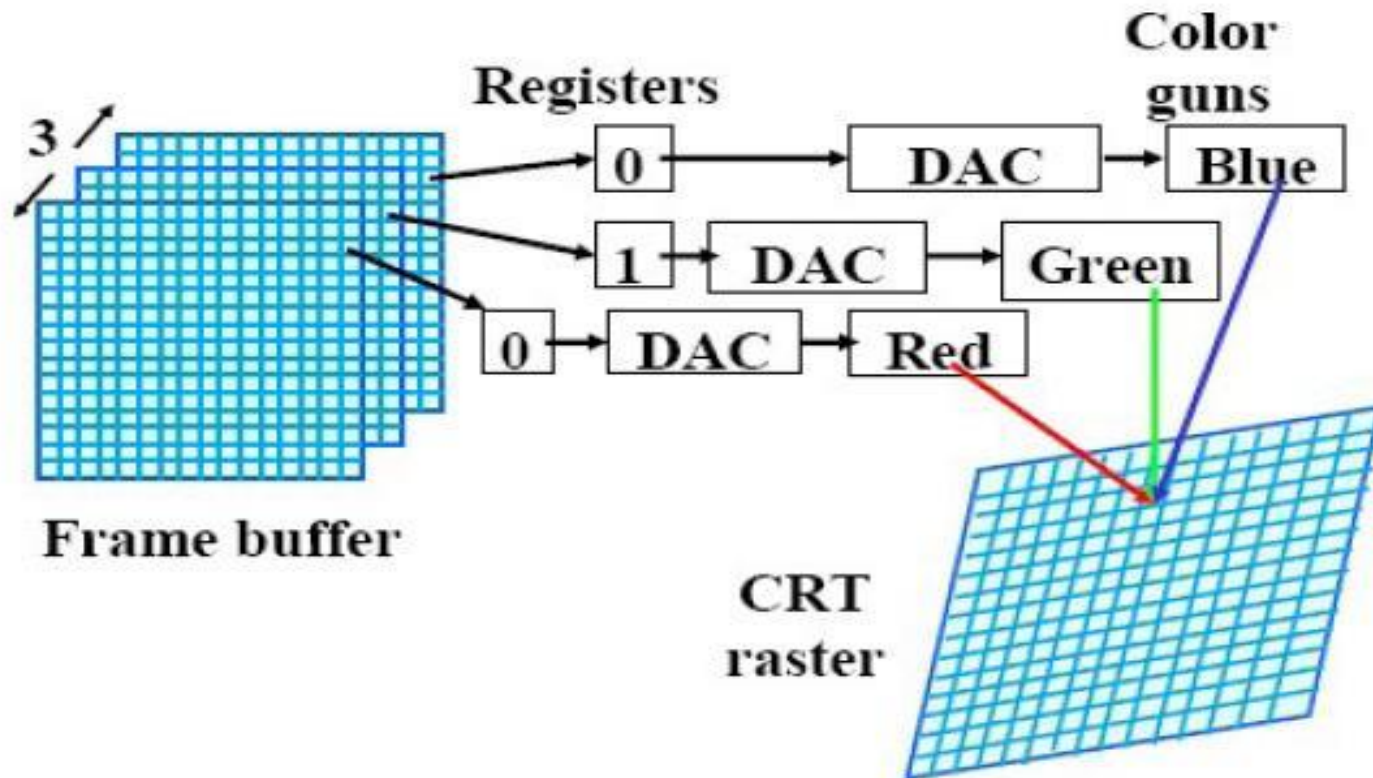
Frame buffer

- A frame buffer is a dedicated block of memory that stores pixel data for an image to be displayed on a screen. It acts as a digital canvas where the graphics rendering process occurs before the image is output to a display device.
- The primary purpose of a frame buffer is to hold the color values for each pixel in the image, allowing the computer to create, manipulate, and display graphics efficiently.
- Color information is typically stored in color channels such as red, green, and blue (RGB), with additional channels like alpha for transparency.

Frame buffer



Frame buffer

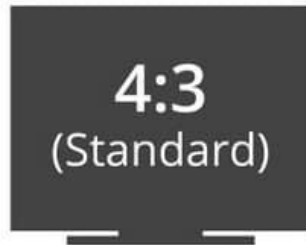


Aspect ratio

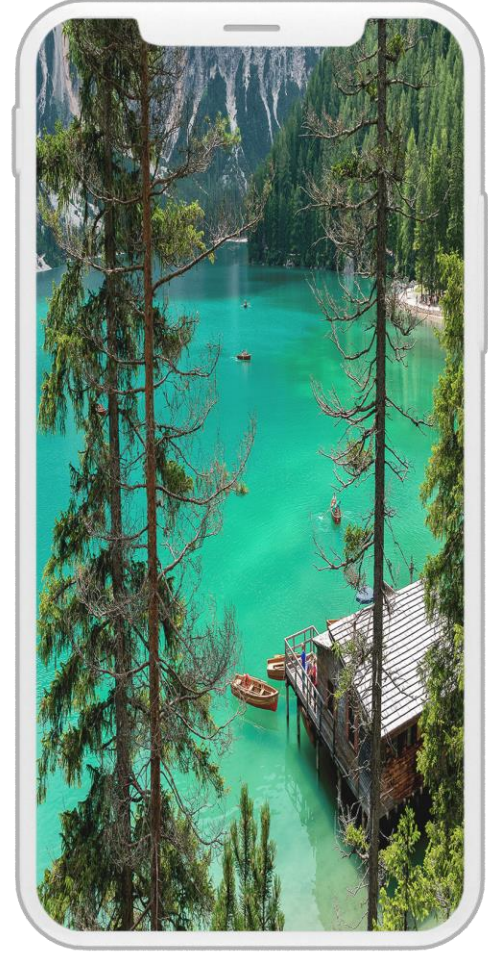
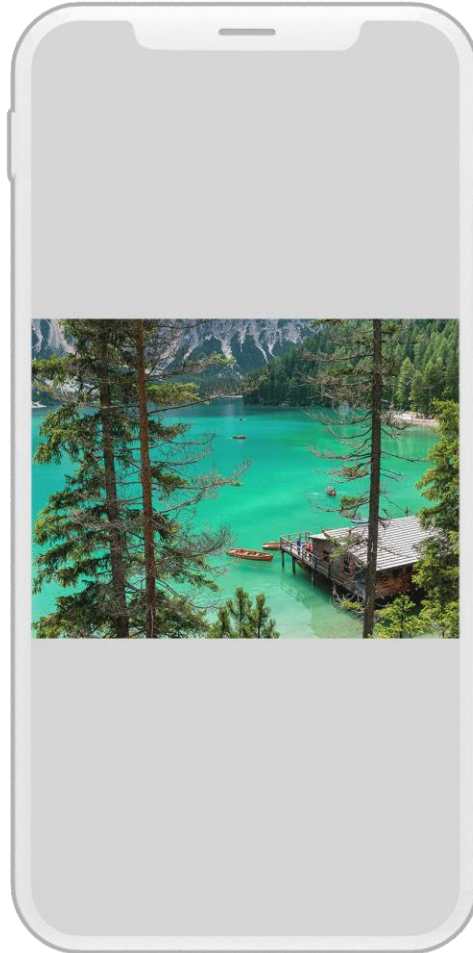
- The aspect ratio dictates the proportional relationship between an image's width and height.
- It's like describing the shape of a picture, and we express it as a ratio, such as 16:9 or 4:3.
- This numeric figure tells us how much wider or taller the image is and significantly influences how we perceive visual content.
- **4:3**: Standard for older monitors and traditional television screens.
- **16:9**: The most common ratio for widescreen displays, HDTVs, and modern monitors.
- **21:9**: Used for ultra-wide monitors and cinematic displays, providing a broader field of view.
- **1:1**: Square ratio, often used in social media & specific artistic

Aspect ratio

- Ensures proper display of content across different devices and screens without distortion.



Aspect ratio

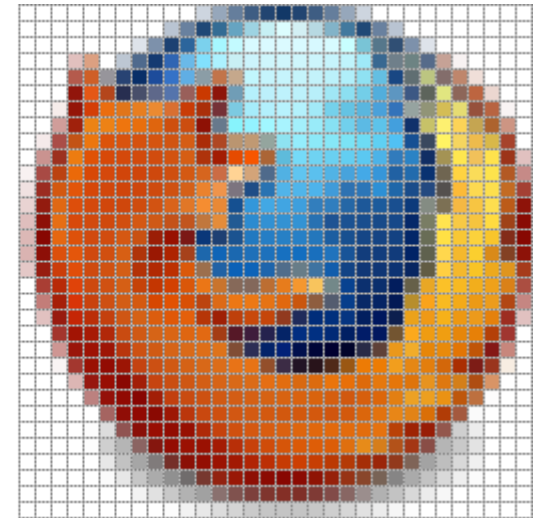
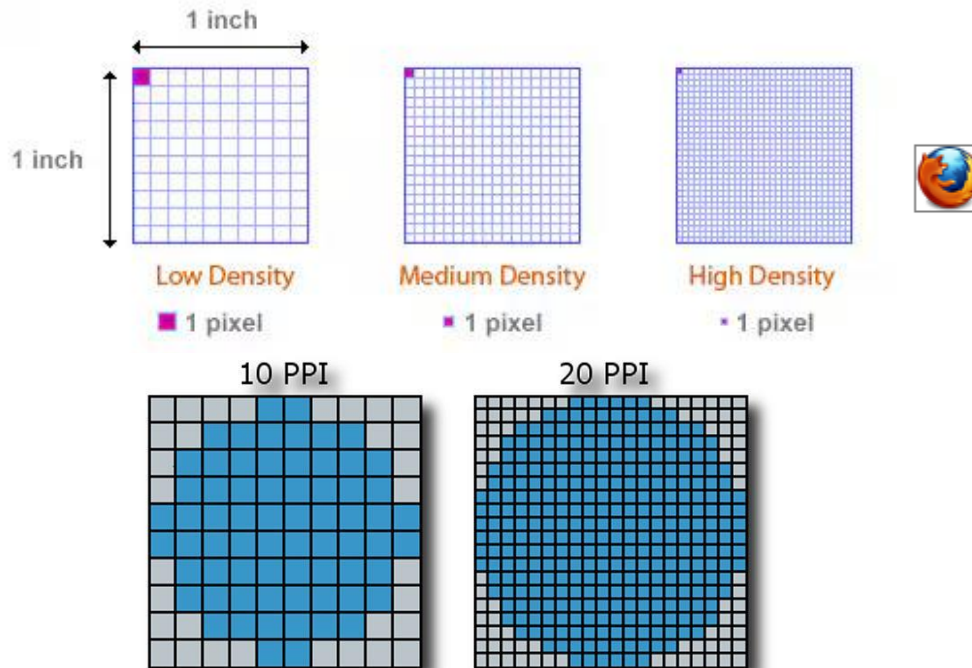


Pixels

- If you hold a magnifier close enough to your computer screen with a picture in it, you can see that the picture is a set of small square dots called pixels (picture element).
- In each pixel, the three fundamental colors red, green and blue (RGB) are mixed in different proportion to create many different colors.
- An image that is presented on the computer screen is made up of pixels.
- The screen consists of a rectangular grid of pixels, arranged in rows and columns. The pixels are small enough that they are not easy to see individually

Pixels

- The full form of the pixel is "Picture Element. A screen is made up of a matrix of thousands or millions of pixels. A pixel is represented with a dot or a square on a computer screen.



Resolution

- Refers to the number of pixels in an image, indicating its detail and clarity.
- More pixels mean higher resolution and potentially better image quality, but other factors like pixel density (PPI - pixels per inch) and viewing distance also impact perceived quality.
- Higher resolution requires more processing power and memory, impacting performance and resource consumption.
- Image resolution is typically measured in pixels per inch (PPI) or dots per inch (DPI). This measurement indicates how many pixels are contained within a linear inch of the image
- Higher PPI values mean more pixels are packed into each inch, resulting in sharper and more detailed images.

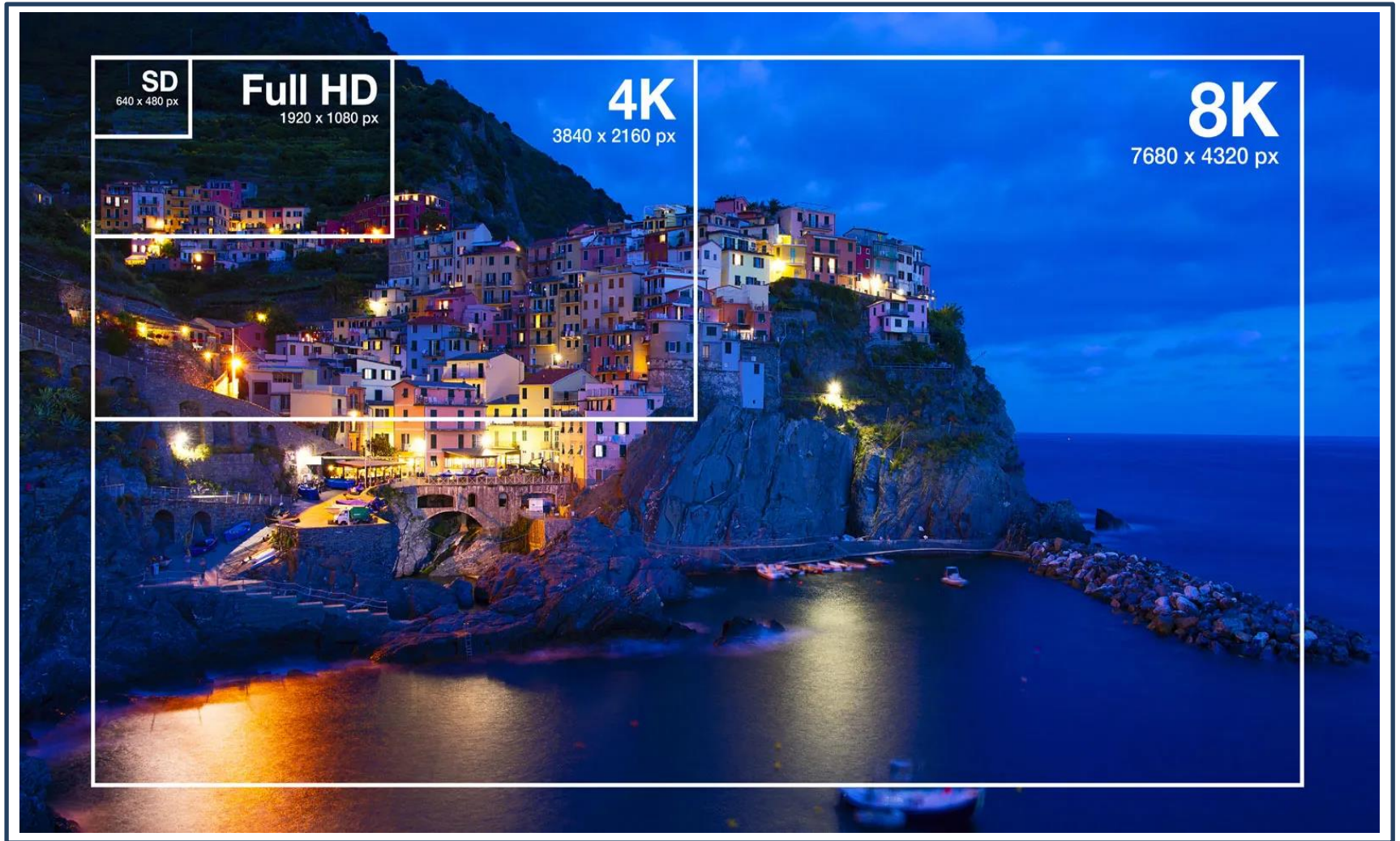
Resolution

- **Factors Affecting Image Resolution:**
- **Pixel Density:** The total number of pixels in an image affects its clarity. An image with dimensions of 1920x1080 has over 2 million pixels, providing a high level of detail.
- **File Size:** Higher resolutions lead to larger file sizes because more pixel data must be stored. This can impact loading times and storage requirements.
- **HD (High Definition):** 1280x720
- **Full HD:** 1920x1080
- **4K Ultra HD:** 3840x2160
- **8K Ultra HD:** 7680x4320

Resolution

Resolution	Number of Pixels	Aspect Ratio
320*200	64000	8:5
640*480	307200	4:3
800*600	480000	4:3
1024*768	786432	4:3
1280*1024	1310720	5:4
1600*1200	1920000	4:3

Resolution



Advantages of interactive graphics

- An interactive graphic is a way to present data to users who visit a page containing animations and customizations, creating a **unique experience for those who wish to check specific information.**
- In interactive Computer Graphics user have some controls over the picture, i.e., the user can make any change in the produced image. One example of it is the ping-pong game.
- Instead of just presenting a fixed frame, you can let each user interact with the image in any way they want.

Advantages of interactive graphics

- Higher Quality
- More precise results or products
- Greater Productivity
- Lower analysis and design cost
- Significantly enhances our ability to understand data and to perceive trends.
- One of the most natural means of communicating with a computer.
- A picture is worth ten thousand words.
- A MOVING picture is worth ten thousand STATIC ones.
- eg. movie, motion dynamics (flight simulators)
- Graphics User Interfaces

Representative uses of cg

- **Entertainment:**
- **Movies:** Special effects, animation, CGI (Computer-Generated Imagery).
- **Video Games:** Real-time rendering, character modeling, environments.
- **Education:**
- **E-learning:** Interactive simulations, educational games.
- **Visual Aids:** Diagrams, illustrations, 3D models.
- **Healthcare:**
- **Medical Imaging:** MRI, CT scans, visualizations of medical data.
- **Surgical Simulations:** Training tools for surgeons.

Representative uses of cg

- **Engineering and Architecture:**
- **CAD (Computer-Aided Design):** Drafting, modeling, and visualization of buildings and mechanical parts.
- **Simulations:** Structural analysis, stress testing.
- **Scientific Research:**
- **Data Visualization:** Graphs, charts, and interactive plots.
- **Simulations:** Climate models, molecular modeling.
- **Business and Marketing:**
- **Infographics:** Visual representation of data and statistics.
- **Product Visualization:** 3D models of products for advertisements.

Classification of hardware & Software

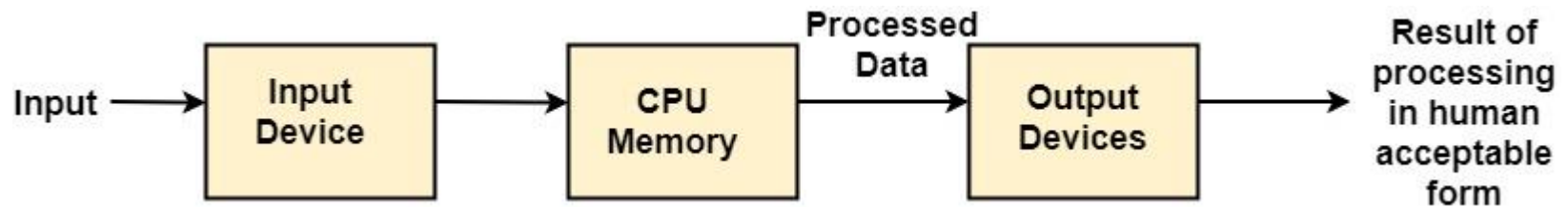
- Computer graphics heavily relies on specialized hardware to achieve real-time performance and high-quality visuals.
- Computer graphics encompass a wide range of hardware and software components that work together to create, manipulate, and display visual content.
- **Processing Units**
- **Central Processing Unit (CPU):**
- General-purpose processor that handles overall system operations and complex computations.
- Collaborates with the GPU in rendering tasks.
- Examples: Intel Core, AMD Ryzen.

Classification of hardware & Software

- **Graphics Processing Unit (GPU):**
 - Specialized processor designed for rendering graphics.
 - Performs parallel processing, making it ideal for handling complex graphics tasks rendering images, and video processing.
 - Examples: Nvidia GeForce, AMD Radeon.
- **Input Devices:**
 - **Mouse:** Essential for interaction and manipulation of graphics.
 - **Graphics Tablet:** Provides precise control for drawing and design.
 - **Digitizer:** Captures real-world images for digital processing.
 - **Scanner:** Converts physical images into digital format.

Classification of hardware & Software

- **3D Mice:** Provide advanced control for 3D modeling.



Classification of hardware & Software

- **Trackball:** It is a pointing device. It is similar to a mouse. This is mainly used in notebook or laptop computer, instead of a mouse.
- This is a ball which is half inserted, and by changing fingers on the ball, the pointer can be moved.
- **Spaceball:** It is similar to trackball, but it can move in six directions where trackball can move in two directions only.
- It is used for three-dimensional positioning of the object.
- **Joystick:** A joystick is a hand-held input device used to control the movement or actions of a digital object on a computer.

Classification of hardware & Software

- **Output Devices:**
- **Monitor:** Displays the generated graphics Type: **LCD, LED, OLED** (Organic light-emitting diodes)
- **Printer:** Produces hard copies of images.
- **Projector:** Displays graphics on a larger screen.
- **Plotter:** Used for printing vector graphics on large formats.
- **VR Headsets:** Provides immersive virtual reality experiences.
- **Memory:**
- **RAM (Random Access Memory):** Temporary storage for active processes and data.
- **VRAM (Video RAM):** Dedicated memory on the GPU for storing textures and frame buffers **Ex:** Graphics Double Data Rate 6 (**GDDR6**)

Classification of hardware & Software

- **Storage Devices:**
- **Hard Disk Drives (HDD):** Traditional storage devices with larger capacity.
- **Solid State Drives (SSD):** Faster storage devices for quicker access to data.
- **Motherboard:** Main circuit board connecting all components, including CPU, GPU, memory, and storage.
- **Power Supply Unit (PSU):** Provides power to all components in the computer.
- **Cooling Systems:**
- **Air Cooling:** Fans and heat sinks to dissipate heat.
- **Liquid Cooling:** Advanced cooling for high-performance systems.

Classification of hardware & Software

- **Software for Computer Graphics:**
- Graphics APIs (Application Programming Interfaces):
- Provide a set of tools for developers to create graphics applications.
- **OpenGL:** Cross-platform API for 2D and 3D graphics.
- **DirectX:** Collection of APIs for handling tasks related to multimedia, especially game programming and video, on Microsoft platforms.
- **Vulkan:** Low-overhead, cross-platform API for high-performance graphics.
- **Graphics Libraries and Frameworks:**
- Provide pre-built functions and routines for graphics tasks.

Classification of hardware & Software

- **WebGL:** JavaScript API for rendering 3D graphics within web browsers.
- **Three.js:** JavaScript library that simplifies the use of WebGL.
- **Modeling and Animation Software:**
 - Used for creating 3D models, animations, and simulations.
- **Blender:** Open-source 3D modeling and animation software.
- **Autodesk Maya:** Professional 3D modeling and animation software.
- **3ds Max:** 3D modeling, animation, and rendering software.
- **Image Editing Software:**
 - Used for creating and editing images and textures.
- **Adobe Photoshop: (Raster Graphics):** Used primarily for photo editing and raster image creation.

Classification of hardware & Software

- **Adobe Illustrator (Vector Graphics):** Focuses on creating vector-based artwork suitable for logos and illustrations.
- **GIMP:** Open-source image editor.
- **Game Engines:**
 - Frameworks for developing video games, incorporating graphics rendering, physics, and more.
 - **Unity:** Popular game engine with extensive graphics capabilities.
 - **Unreal Engine:** Advanced game engine known for high-quality graphics and real-time rendering.
- **Visualization Software:**
 - Used for scientific and data visualization.

Classification of hardware & Software

- **Matplotlib:** Python library for creating static, animated, and interactive visualizations.
- **ParaView:** Open-source software for data analysis and visualization.
- **Rendering Software:**
 - Specializes in converting 3D models into high-quality images.
- **RenderMan:** Rendering software developed by Pixar.
- **V-Ray:** Rendering engine used in architecture, film, and visual effects.
- **CAD (Computer-Aided Design) Software:**
 - Used for designing and modeling engineering and architectural projects.
- **AutoCAD:** Widely used CAD software.

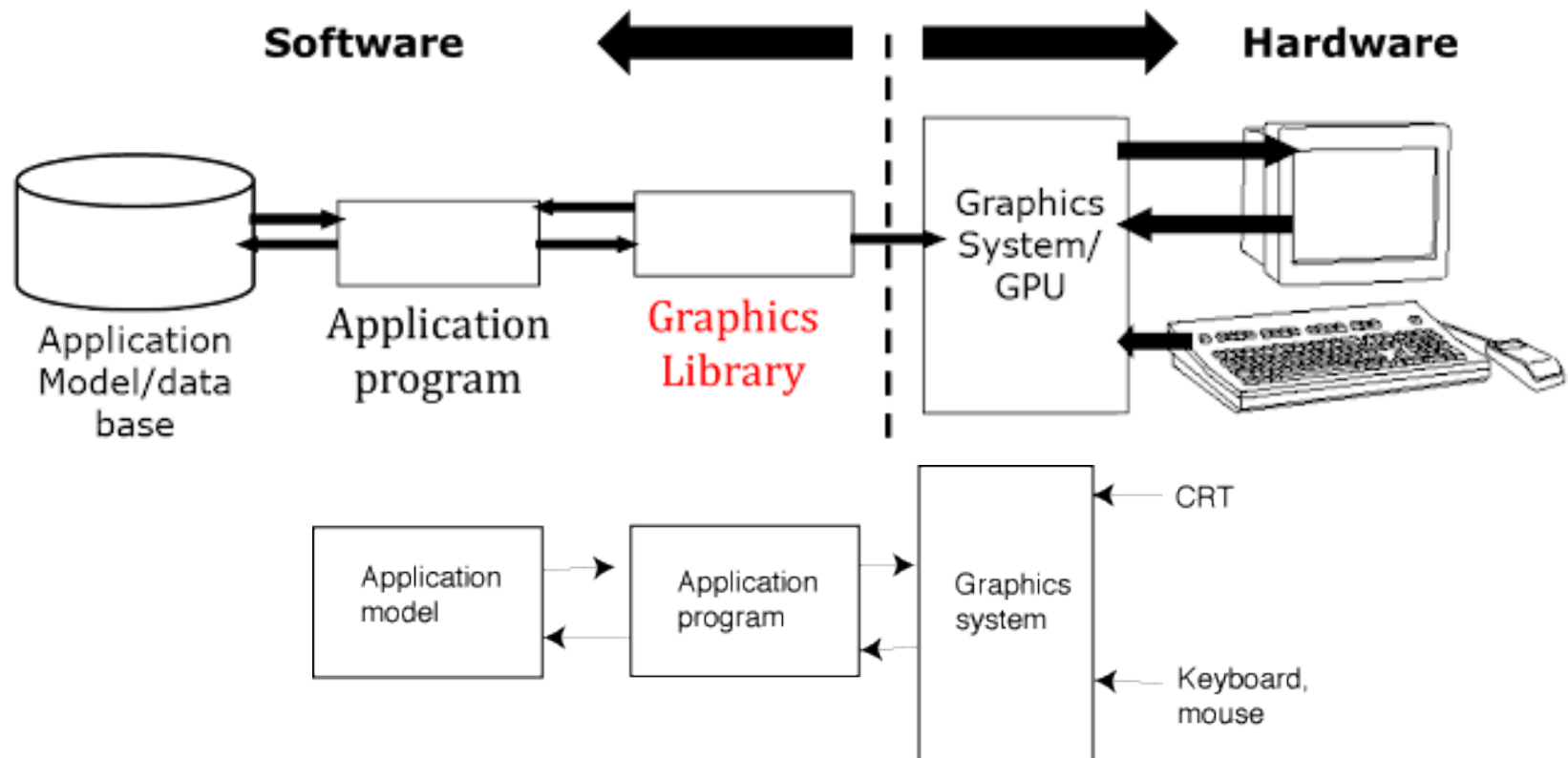
Classification of hardware & Software

- **SolidWorks:** 3D CAD design and modeling software.
- **Video Editing Software:**
 - Used for editing and producing video content.
 - **Adobe Premiere Pro:** Professional video editing software
 - **Final Cut Pro:** Video editing software for Mac users.
- **Simulation Software:**
 - Used for creating simulations and visual effects.
 - **Houdini:** Software known for its capabilities in simulation and visual effects.
- **Simulink:** Used for modeling, simulating, and analyzing multidomain dynamic systems.
- Hardware and software are essential in computer graphics, each contributing to the creation, manipulation, and display.

Conceptual framework for interactive graphics

- **Conceptual Framework has the following elements:**
- **Graphics Library** – Between application and display hardware there is graphics library / API.
- **Application Program** – An application program maps all application objects to images by invoking graphics.
- **Graphics System** – An interface that interacts between Graphics library and Hardware.
- Modifications to images are the result of user interaction
- **Interactive graphics involve creating visual representations that respond dynamically to user input.**

Conceptual framework for interactive graphics



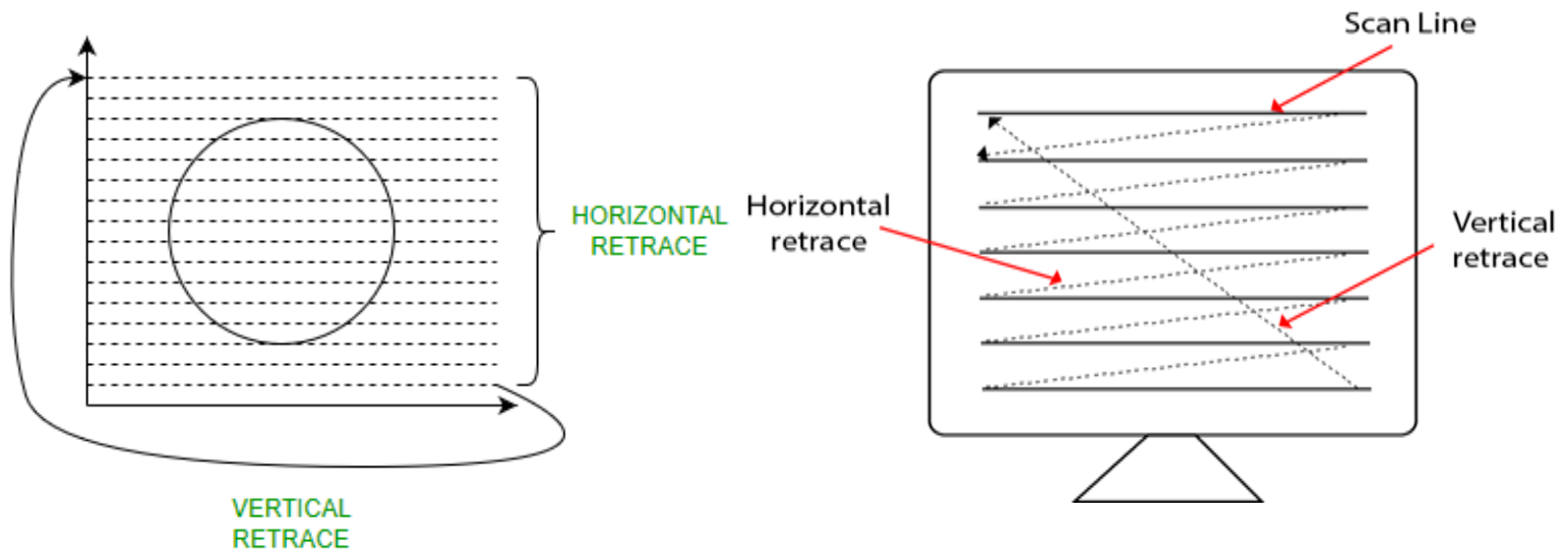
Conceptual framework for interactive graphics

Raster Scan Display System

- A **raster scan display system** is a method used in computer graphics to display images on a screen. It is the most common method used in devices like CRT monitors, LCDs, and plasma screens.
- **Raster Scan Displays** are most common type of graphics monitor which employs CRT. It is based on television technology. In raster scan system electron beam sweeps across the screen, from top to bottom covering one row at a time.
- A pattern of illuminated (shining) pattern of spots is created by turning beam intensity on and off as it moves across each row. A memory area called refresh buffer or frame buffer stores picture definition. This memory area holds intensity values for all screen points.

Raster Scan Display System

- In raster scan systems refreshing is done at a rate of 60-80 frames per second. Refresh rates are also sometimes described in units of cycles per second / Hertz (Hz).



Raster Scan Display System

Random Scan	Raster Scan
1. It has high Resolution	1. Its resolution is low.
2. It is more expensive	2. It is less expensive
3. Any modification if needed is easy	3.Modification is tough
4. Solid pattern is tough to fill	4.Solid pattern is easy to fill
5. Refresh rate depends or resolution	5. Refresh rate does not depend on the picture.
6. Only screen with view on an area is displayed.	6. Whole screen is scanned.
7. Beam Penetration technology come under it.	7. Shadow mark technology came under this.
8. It does not use interlacing method.	8. It uses interlacing
9. It is restricted to line drawing applications	9. It is suitable for realistic display.

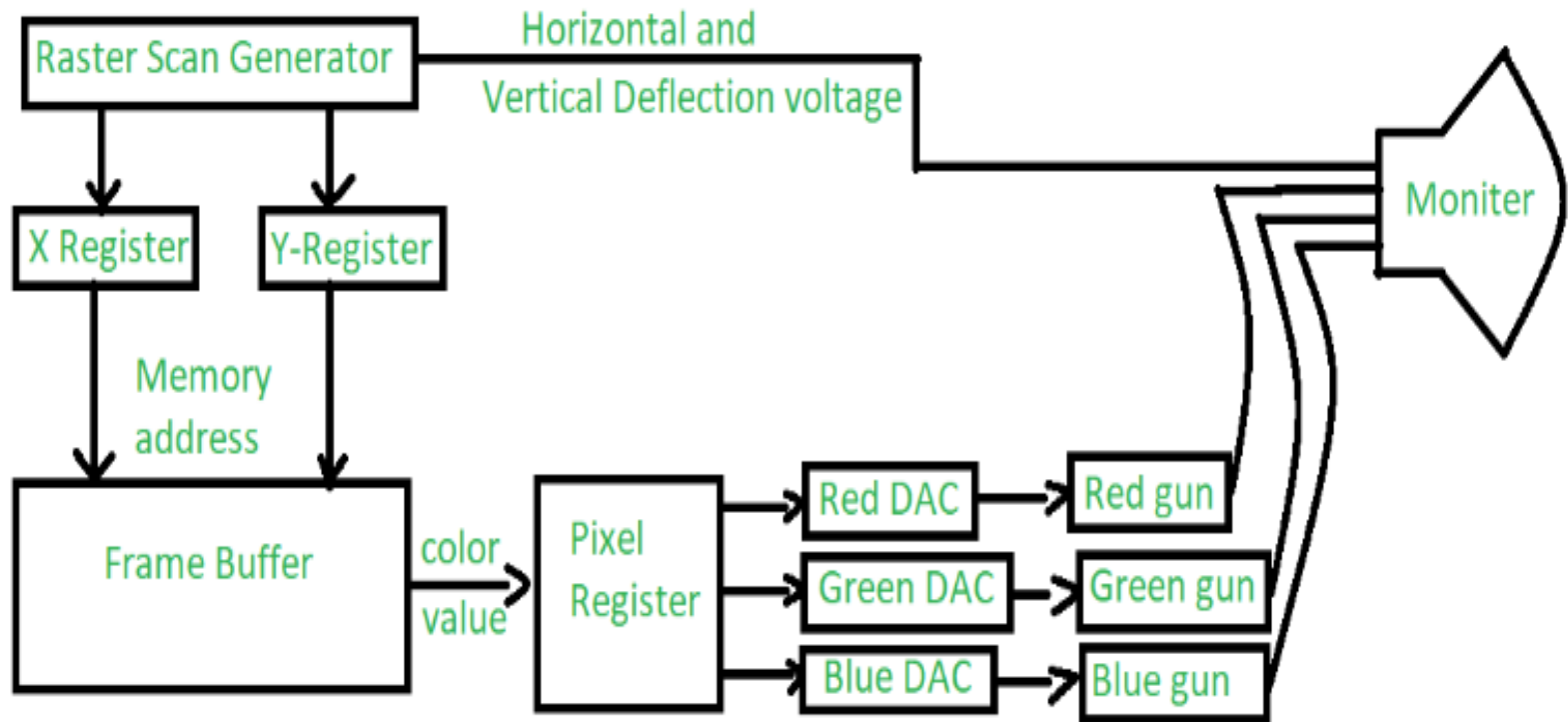
Video Controller

- A video controller, also known as a display controller or graphics card, is a crucial component in computer graphics that bridges the gap between the computer's internal data and the visual output displayed on a monitor or screen.
- The video controller ensures that the data in the computer's memory is correctly converted into visual information and displayed on the screen.
- **Functions of a Video Controller:**
- **Generates video signals:** Converts digital image data into analog signals that can be understood by display devices.

Video Controller

- **Controls display timing:** Manages the refresh rate, synchronization signals, and other timing parameters for the display.
- **Manages frame buffer:** Accesses and controls the memory area (frame buffer) where image data is stored.
- **Handles color processing:** Converts color data into appropriate signals for the display.
- **Accelerates graphics processing:** Many modern video controllers incorporate dedicated graphics processing units (GPUs) to handle complex calculations and image manipulations efficiently.
- **Multiple Display Management:** allowing users to connect several monitors to a single computer.

Video Controller

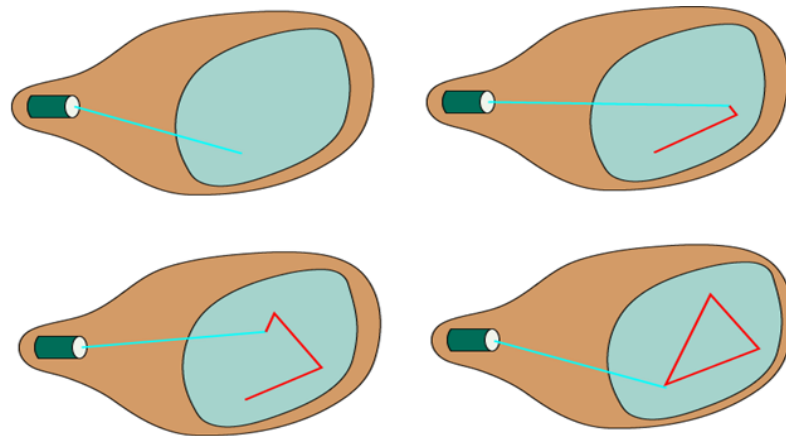


Random Scan Display Processor

- Input in the form of an application program is stored in the system memory along with graphics package. Graphics package translates the graphic commands in application program into a display file stored in system memory.
- This display file is then accessed by the display processor to refresh the screen. The display processor cycles through each command in the display file program. Sometimes the display processor in a random-scan is referred as Display Processing Unit / Graphics Controller.
- **Random Scan Display:**

Random Scan Display Processor

- Random Scan System uses an electron beam which operates like a pencil to create a line image on the CRT screen. The picture is constructed out of a sequence of straight-line segments.
- Each line segment is drawn on the screen by directing the beam to move from one point on the screen to the next, where its x & y coordinates define each point.



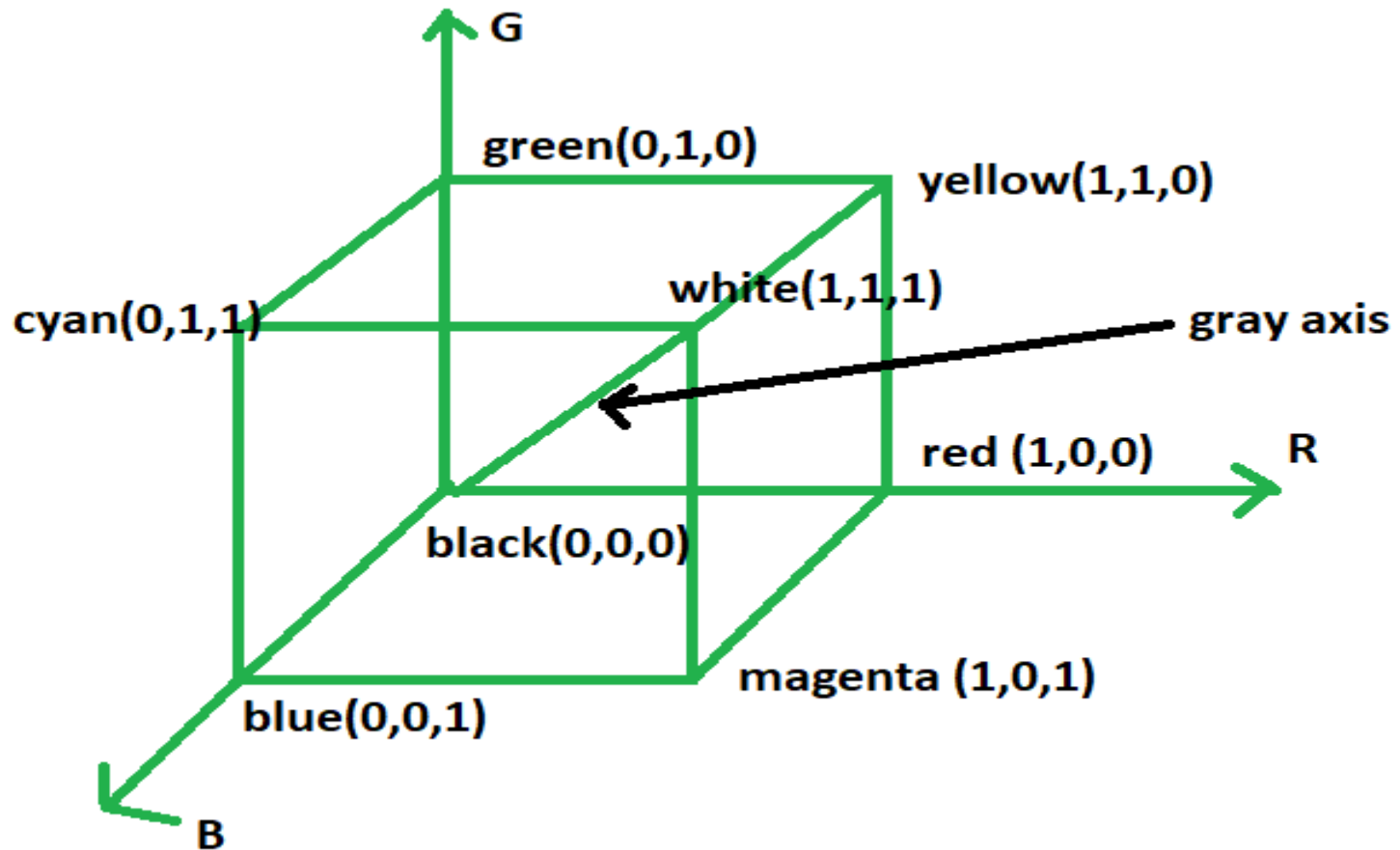
Color Model

- Different types of colour models are used in multiple fields like in hardware, in multiple applications of creating animation, etc.
- **RGB (Red, Green, Blue)**
- Used for digital screens by combining red, green, and blue light.
Application: Computer monitors, TVs, web design, digital cameras.
- **CMYK (Cyan, Magenta, Yellow, Key/Black)**
- A subtractive color model used in color printing.
Application: Printers, publishing, brochures, posters.

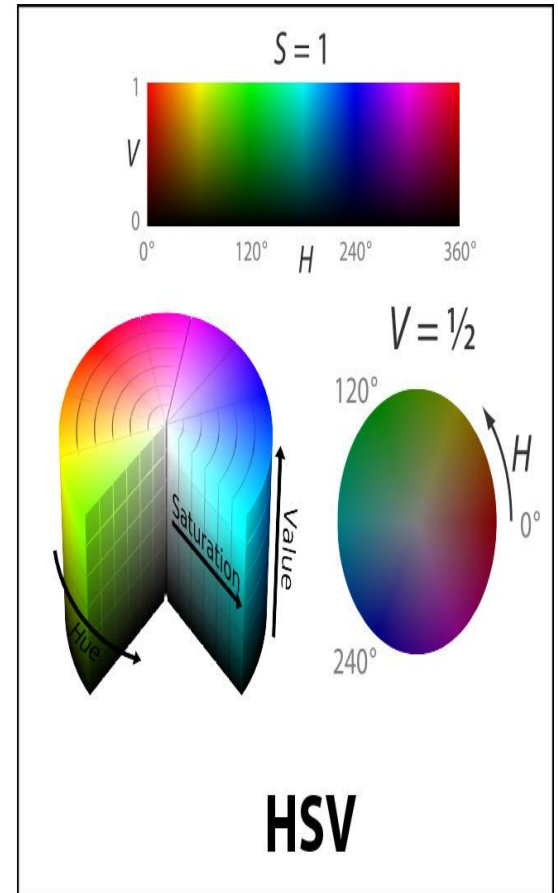
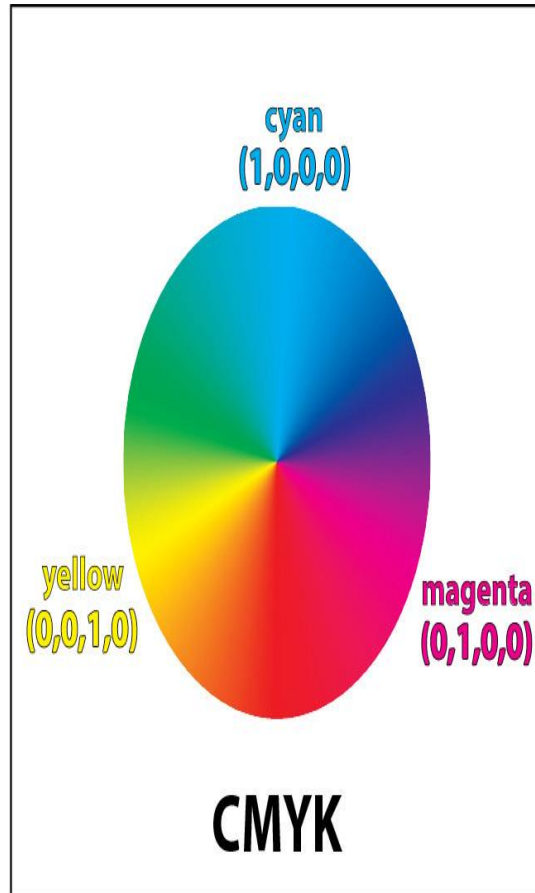
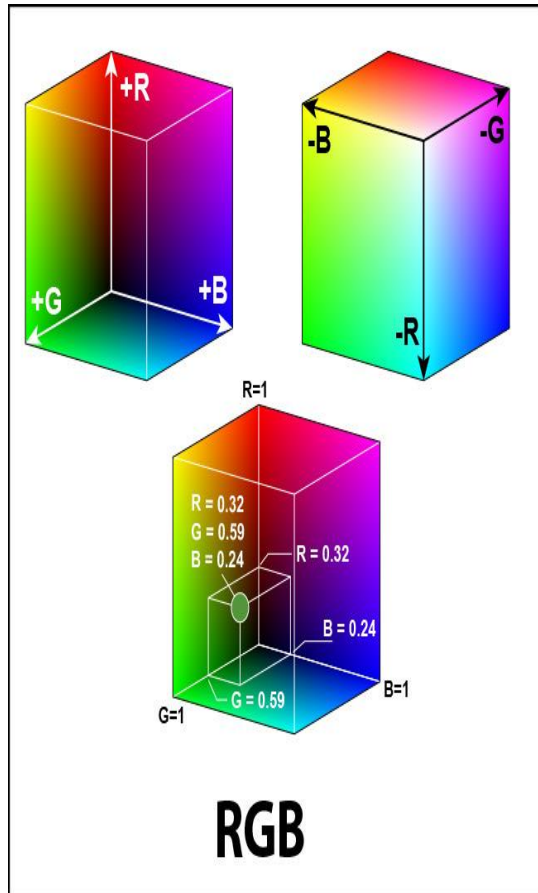
Color Model

- **HSV (Hue, Saturation, Value)**
 - Represents colors based on how humans perceive them.
 - **Application:** Image editing tools, color pickers in design software.
- **YIQ (Luminance, In-phase, Quadrature)**
 - Used in NTSC analog television systems to separate brightness and color.
 - **Application:** U.S. analog TV broadcasting, image compression.

Color Model

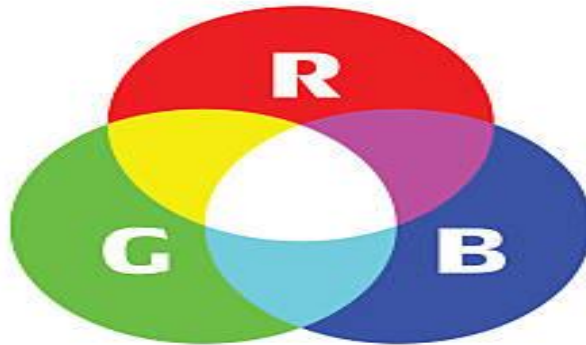


Color Model



Color Model

Color Models

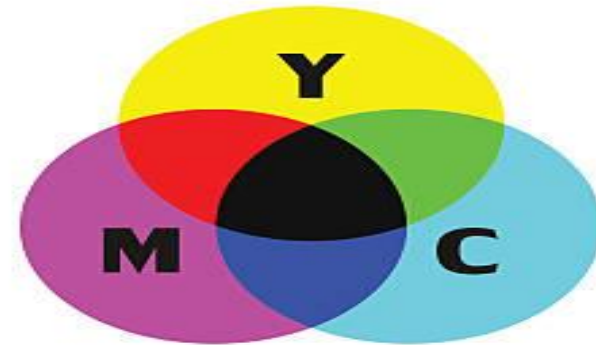


RGB

Red Green Blue



RGB Color Wheel



CMYK

Cyan Magenta Yellow Key color



HSL (Hue-Saturation-Lightness)



HSV (Hue-Saturation-Value)

Types of Computer Monitor

Types of Computer Monitors



CRT Monitor



LCD Monitor



LED Monitor



OLED Monitor



Curved Monitor



Ultrawide Monitor



4K Monitor



6K Monitor



Gaming Monitor



HDR Monitor



Medical Monitor



Broadcast Monitor



8K Monitor



E-Link Monitor



Security Monitor



Multi-Screen Arrays



Projector

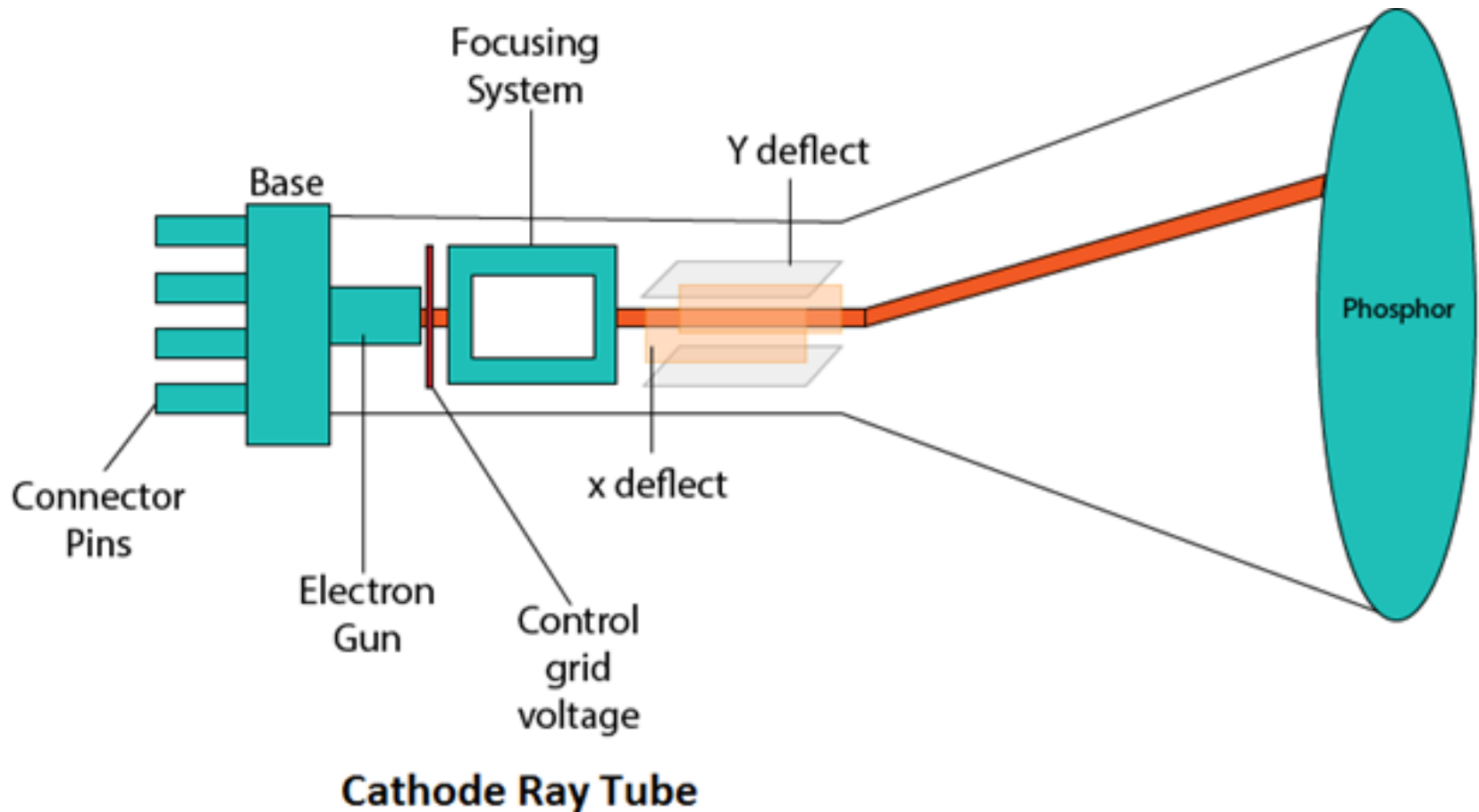


Portable Monitors

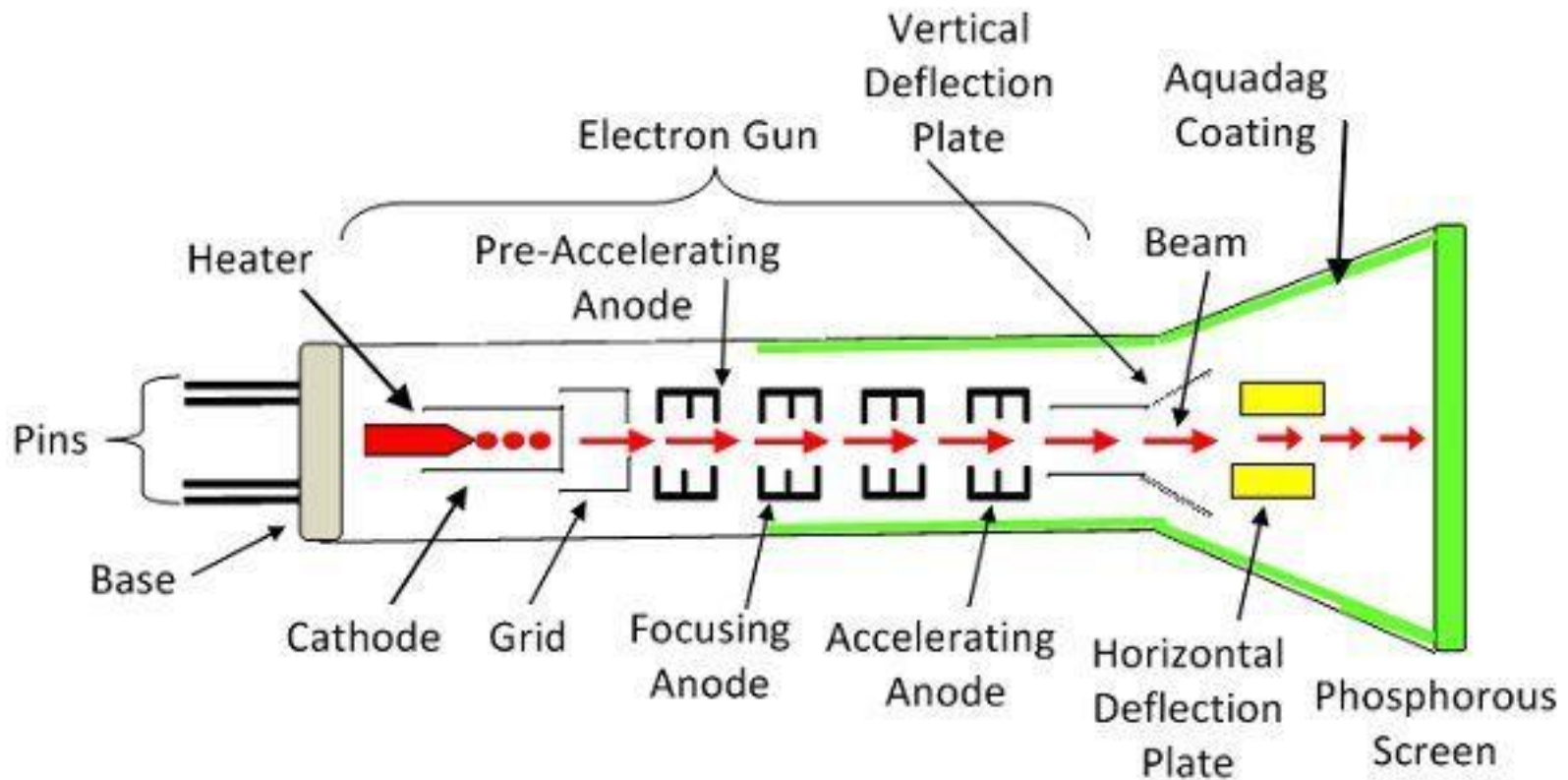


Touchscreen

CRT



CRT



Cathode Ray Tube

Circuit Globe

CRT Component

- **Electron Gun:** Electron gun consisting of a series of elements, primarily a heating filament (heater) and a cathode. The electron gun creates a source of electrons which are focused into a narrow beam directed at the face of the CRT.
- **Control Electrode:** It is used to turn the electron beam on and off.
- **Focusing system:** It is used to create a clear picture by focusing the electrons into a narrow beam.
- **Deflection Yoke:** It is used to control the direction of the electron beam. It creates an electric or magnetic field which will bend the electron beam as it passes through the area.
- **Phosphorus-coated screen:** The inside front surface of every CRT is coated with phosphors.

Color CRT Monitor

- **Beam Penetration Method:** This method is used in random scan monitors and works like a CRT with a single electron gun. It uses two phosphor layers (usually red and green) on the screen's inner surface. By controlling the speed of the electron beam, different colors are produced through penetration of the layers.
- **Shadow Mask Method:** Common in raster scan systems like color TVs, this method uses three electron guns—one each for red, green, and blue. A shadow mask ensures that each gun hits the correct color phosphor dot. It provides a wider range of colors and better image quality.

Color CRT Monitor

- **Beam Penetration Method:** The beam penetration method for displaying color pictures has been used with random scan monitors. Two layers of phosphor, usually red and green are coated onto the screen and the displayed color depends on how far the electron beam penetrates into the phosphor layers.
- A **beam of slow electrons** excites only the outer red layer, fast electrons penetrate through red layer and excite the inner green layer. The electrons with intermediate speed produce the combinations of red and green color are emitted to show two additional colors, orange and yellow. The speed of electrons is controlled by the beam-acceleration voltage. But the quality of the pictures is not good here.

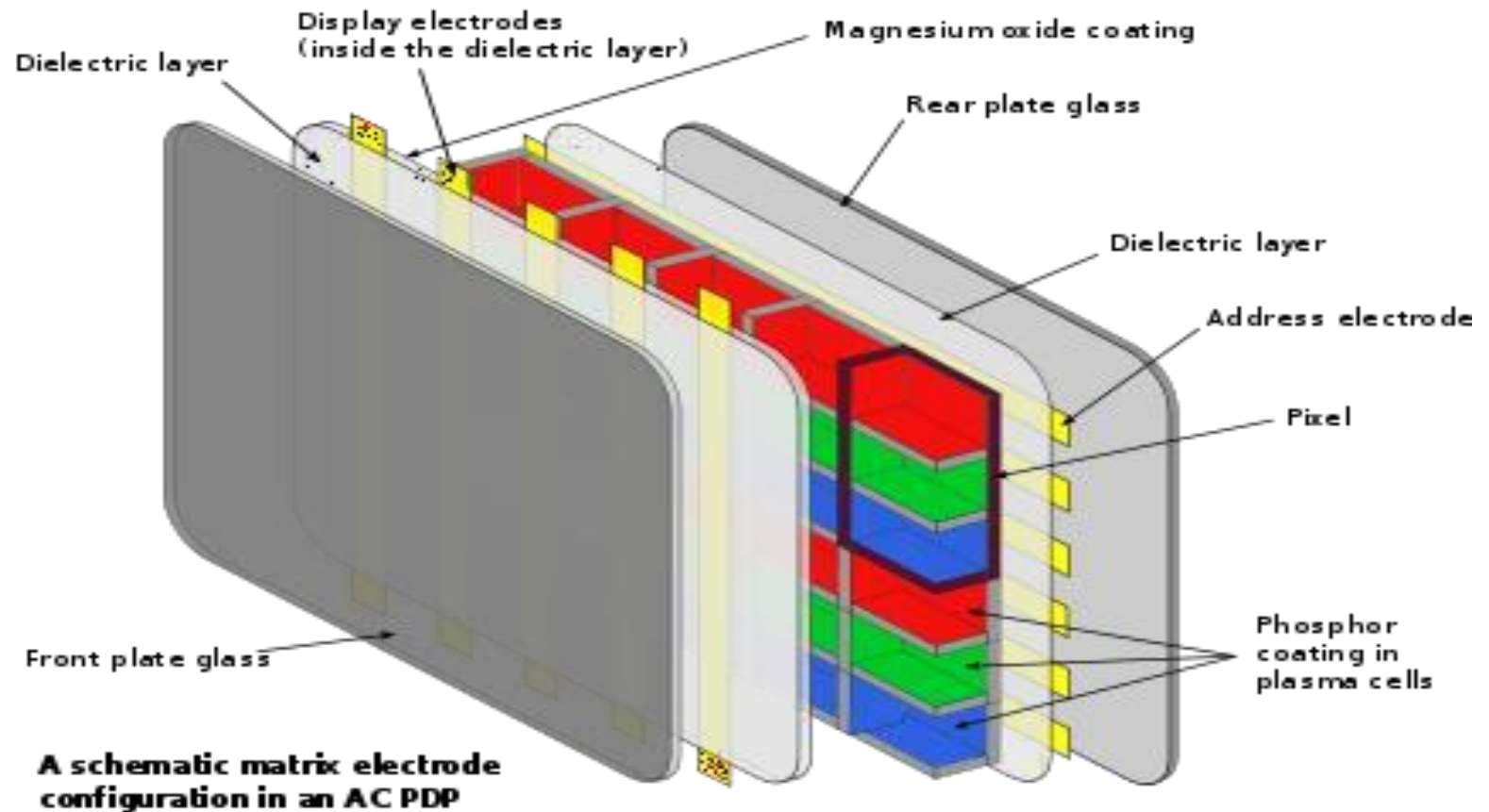
Color CRT Monitor

- **Shadow Mask Method:** It can produce a much wider range of color than the
- beam-penetration method.
- A shadow-mask CRT has three phosphor color dots at each pixel position with red, green and blue.
- This type of CRT has three electron guns, one for each color.
- The three electron beams are deflected and focused as a group onto the shadow mask, which contains corresponding holes for each pixel position.
- The color variation is obtained by varying the intensity levels of the three electron beams.
- For example, turning off the green and red beam, we will get only blue color.

Flat Panel Displays

- It refers to a class of video devices that have reduced volume, weight and power requirements compared to CRT.
- A main advantage of flat-panel displays is that **they are thinner than CRTs** and one can hang them on walls or wear on our wrists.
- The **two categories of flat-panel displays** are
 - **Emissive displays:** the device which convert electrical energy into light. (Plasma panel), Thin-Film Electroluminescent Display, LED.
 - **Non-Emissive displays:** used optical effects to convert sunlight or light from some other source into graphic patterns. (LCD)

Plasma panel



Plasma panel

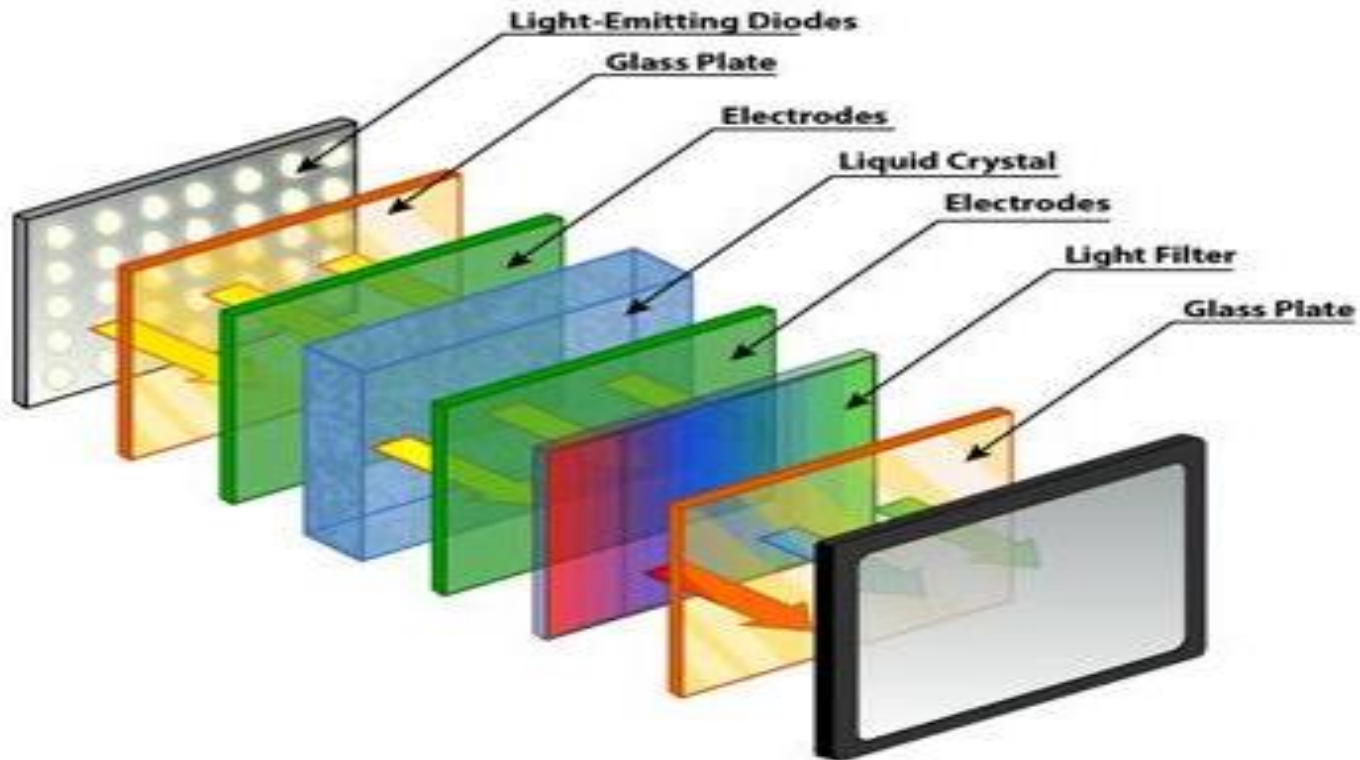
- A **plasma display panel** consists of hundreds of thousands of small luminous cells (or pixels) sandwiched between two glass plates.
- Each cell contains a mixture of inert gases like **neon and xenon**, which turn into plasma and emit **ultraviolet (UV) light** when electrically charged.
- This UV light then excites phosphor material in the cell to produce visible light (red, green, or blue).
- The panel includes **electrodes—address electrodes** on the rear glass and **sustain electrodes** on the front—that control the electrical discharges.

LED

- A matrix of diodes is arranged to form the pixel positions in the display, and the picture definition is stored in a refresh buffer.
- As in scan-line refreshing of a CRT, information is read from the refresh buffer and converted to voltage levels that are applied to the diodes to produce the light patterns in the display.
- LED (light-emitting diode) display
- LED displays are commonly used in various electronic devices, including television screens, computer monitors, digital signage, and outdoor displays.

LED

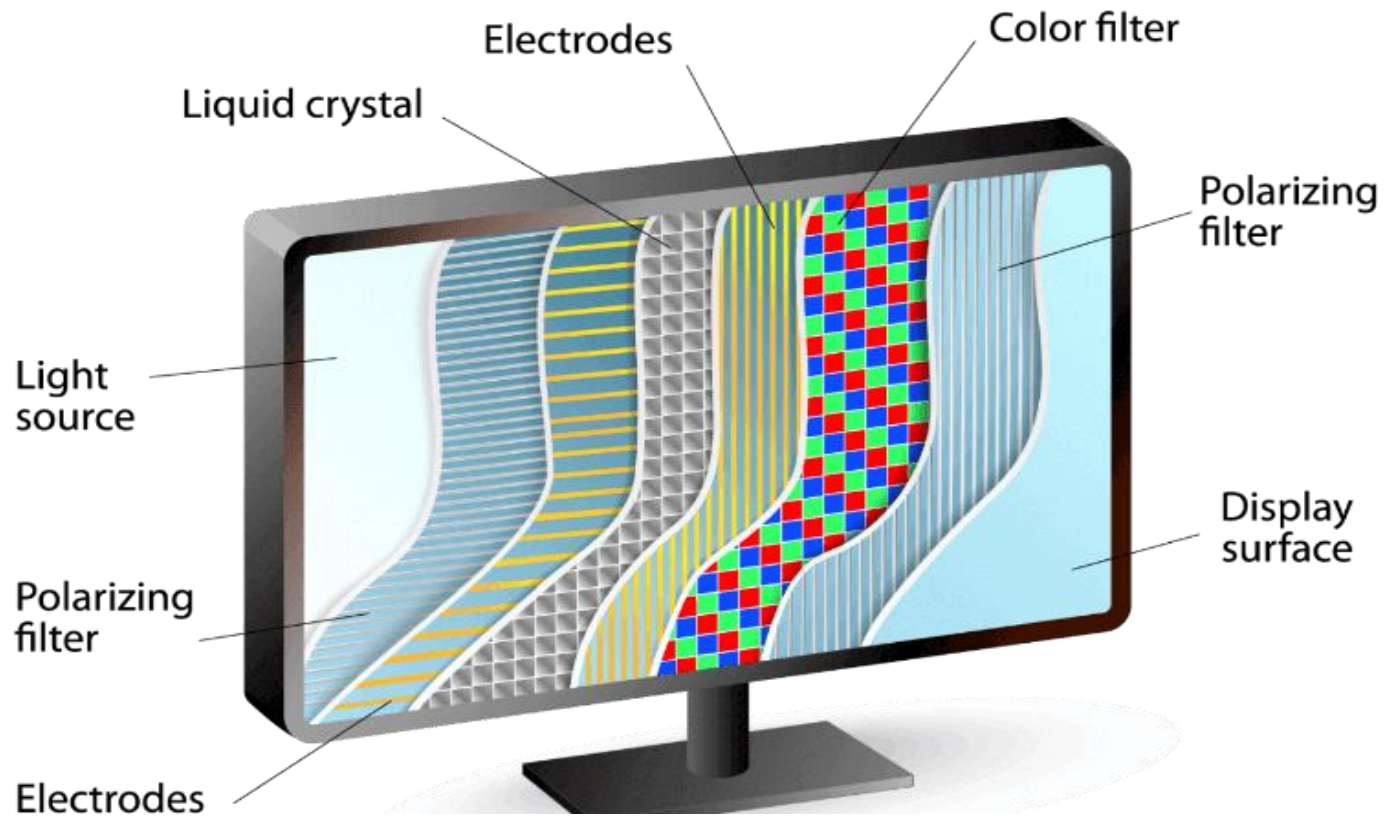
LED Technology Process



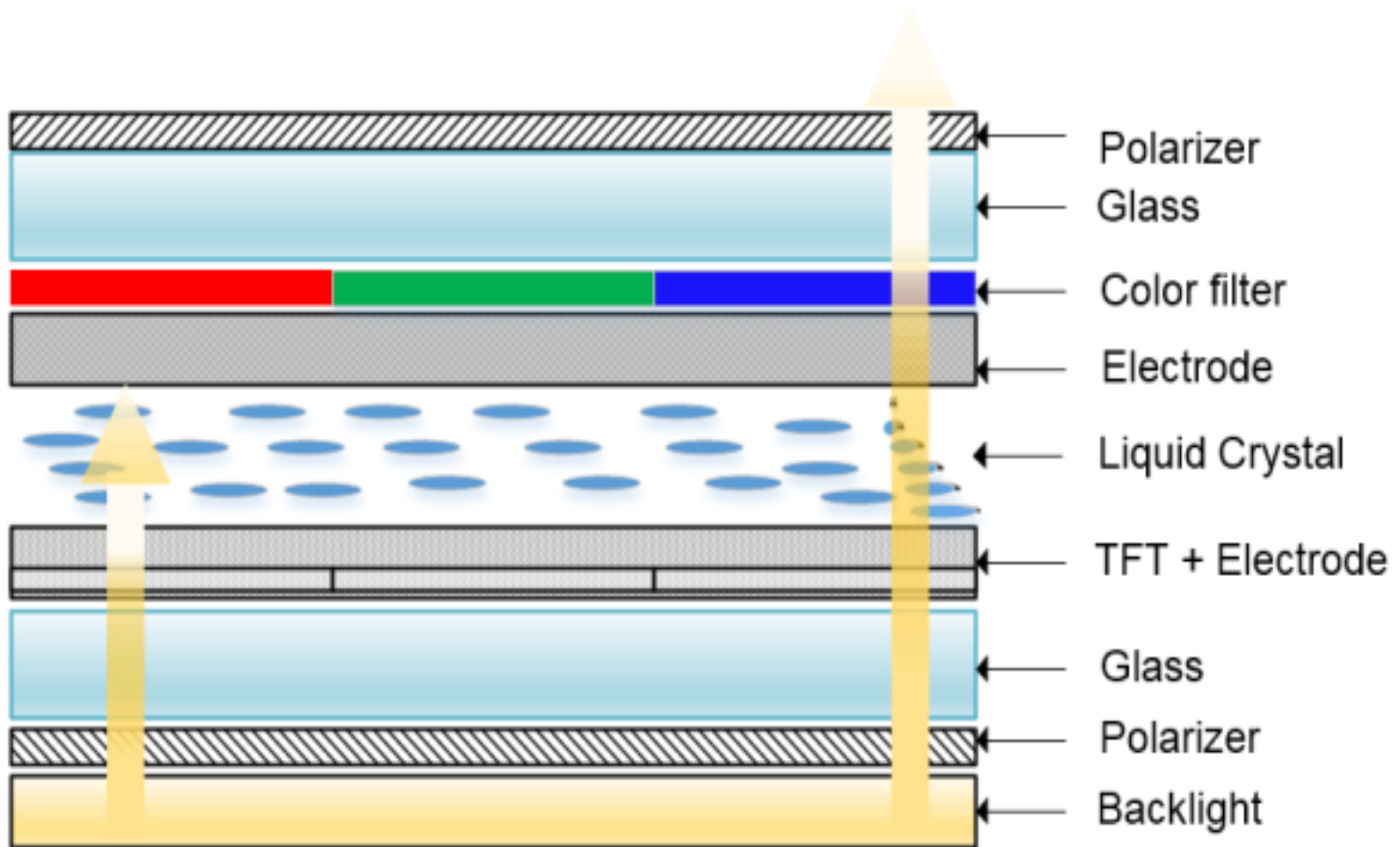
LED light source: shining through liquid crystal cells and glass plates.

LCD

LIQUID CRYSTAL DISPLAY



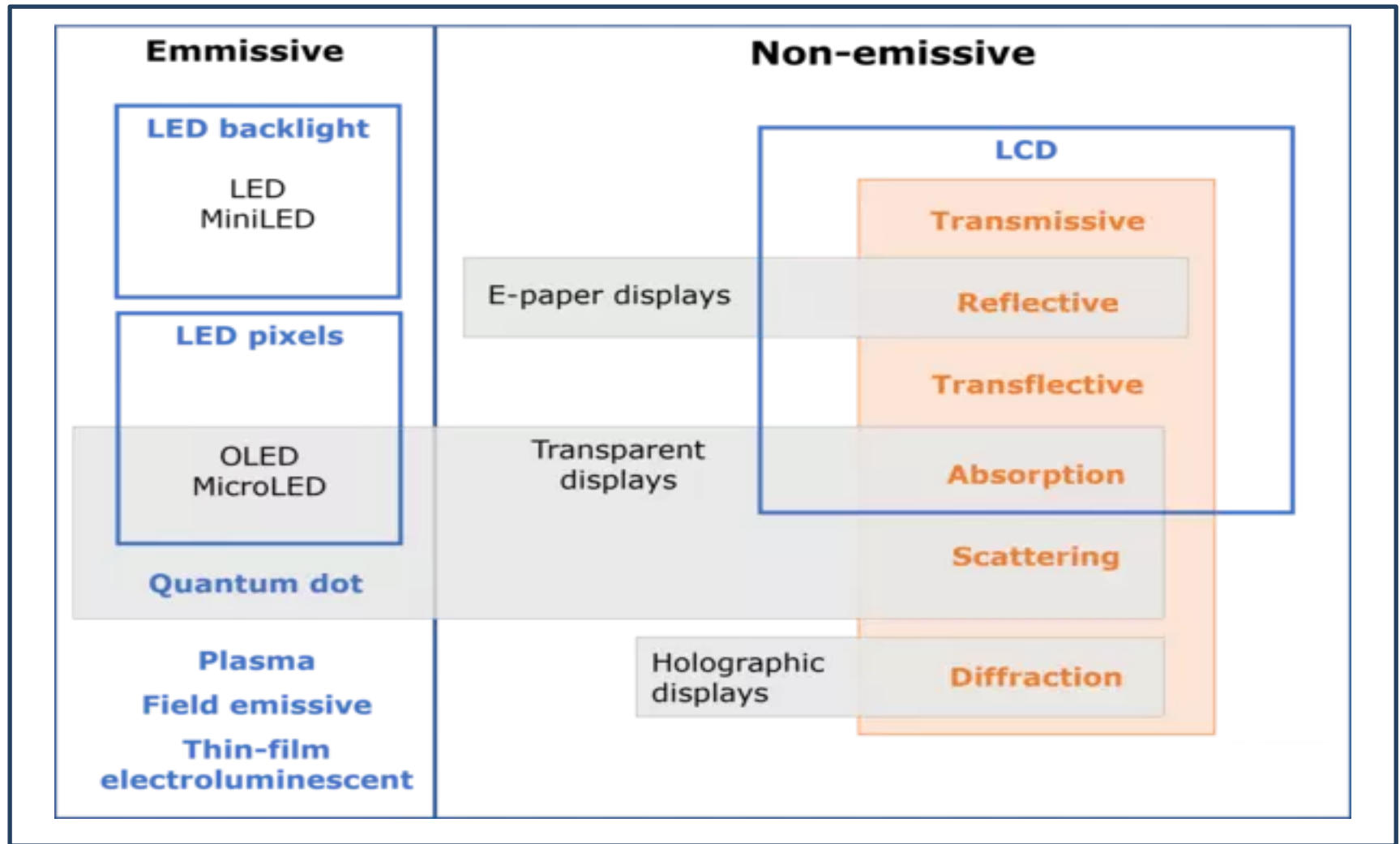
LCD



LCD

- LCD stands for Liquid Crystal Display. It is a flat panel display technology, mainly used in TVs and computer monitors, nowadays it is used for mobile phones also.
- These LCDs are completely different from that old CRT displays, it uses liquid crystals instead of cathode ray in its primary form of operation .
- In LCD display, it consists of millions of pixels made of crystal and arranged in a rectangular grid. In LCD it has backlights that provide light to each pixel.
- Each pixel has a red, green, and blue (RGB) sub-pixel that can be turned on or off.

Emmissive Non-Emissive device



Interactive device

- Interactive devices are broadly categorized into Input Devices and Output Devices (sometimes hybrid as well).

Device	Description	Example Use
Keyboard	Standard alphanumeric input device.	Typing commands or text in a graphics application.
Mouse	Pointing device to select and manipulate objects.	Drawing lines, selecting tools.
Joystick	Used for 2D/3D movement.	Games, virtual reality navigation.
Trackball	A stationary pointing device.	Used in design where space is limited.
Light Pen	A light-sensitive pen that detects position on screen.	Used to draw or select items on CRT screens.
Graphics Tablet	Pen-based input on a touch-sensitive pad.	Used for precise drawing, e.g., in CAD or digital art.

Interactive device

Device	Description	Example Use
Monitor (CRT/LCD/LED)	Displays 2D or 3D graphics.	Main visual output for all graphics applications.
Printers/Plotters	Produce hard copy of graphics.	Architectural blueprints, engineering designs.
Projectors	Large-scale display of graphics.	Presentations, interactive walls.
VR Headsets	Immersive 3D environments.	Virtual reality applications.

Interactive device

- Interactive Output Devices (output with input features)

Device	Description	Use
Touchscreen Monitor	Acts as both input and output.	Used in POS, ATMs, kiosks.
Interactive Whiteboards	Large touch-sensitive displays.	Classrooms, presentations.
Haptic Devices	Provide force/feedback to simulate touch.	Surgical simulators, VR training.

Thank You